

CMORizer – Documentation

Introduction

CMORizer is a Fortran-based tool designed to extract and/or calculate variables from raw WRF model output files (wrfout, wrfpress, wrfxtrm) and generate CMORized output. The output can be aggregated per year, month, or for a specific period. The tool supports 1-hourly, 3-hourly, and 6-hourly output frequencies. Daily data can be produced afterward using command-line tools such as CDO and NCO.

The code supports multiple map projections (e.g. Lambert Conformal Conic, rotated latitude-longitude) and continues to be actively developed. CMORizer was originally developed by a group led by **Sebastian Knist** and **Klaus G3rgen**. It has since been updated and extended to improve usability and support broader applications.

Repository Access

The CMORizer repository is hosted on a private GitLab server at the following location:
<https://icg4geo.icg.kfa-juelich.de/ExternalRepos/pCMORizer/>

The repository is private, so access must be granted individually. To obtain access:

Send an email to Klaus G3rgen (k.goergen@fz-juelich.de) with:

- Your full name and affiliation
- Your institutional email address
- Your GitLab username

Once access is granted, clone the recommended branch ("josipa"):

```
git clone -b josipa  
https://icg4geo.icg.kfa-juelich.de/ExternalRepos/pCMORizer.git
```

This branch include several enhancements that have been added to the original code by **Josipa Milovac** (milovacj@unican.es):

- Support for different map projections (e.g., **Lambert, rotated lat-lon**)
- New variables added to postprocessing
- Possibility to extract data directly from wrfpress files added
- Functionality to postprocess static data added
- Simplified internal processes for better maintainability and bugs fixed

Setup Instructions

Follow these steps to install and configure CMORizer:

1. Install dependencies:
 - NetCDF-C
 - NetCDF-Fortran
 - Fortran compiler (gfortran or ifort)
 - CDO
 - NCO
 - Python
2. Optional: Use the included conda environment file (environment.yaml):

```
conda env create -f environment.yaml
conda activate cmorizer
```

3. Configure the Makefile:
 - Open the Makefile in a text editor
 - Set the correct Fortran compiler (FC variable)
 - Set your paths to NetCDF-C and NetCDF-Fortran libraries
 - Compile using: make
 - Clean executables using: make veryclean

NOTE: NetCDF must be compiled using the same compiler you plan to use for CMORizer.

4. Configure CMORizer parameters:
 - Edit the files named runctrl.current.nml_template*
 - Adjust domain settings, metadata, projection information, and file paths
 - Set paths to your geo_em files and output directories
5. Set input file paths:
 - Edit the run_pCMORizer_in_quasi_parallel.sh script
 - Set the path to the folder containing wrfout, wrfpress, and wrfxtrm files
 - All input files must be located in the same directory

Running CMORizer

Single variable example:

To process a single variable and verify your setup, run:

```
./run_CMORizer_in_quasi_parallel.sh [YEAR] [DOMAIN] [VARIABLE]
[PROJECT_NAME] [FREQUENCY]
```

Example:

```
./run_CMORizer_in_quasi_parallel.sh 2020 d02 tas EUROCORDEX 1hr
```

- This will create a working project directory
- Raw WRF files will be linked automatically
- A folder for the selected variable will be created

Batch processing:

To process a full list of variables, use:

```
./loop_variables.sh [CSV_FILE] [YEAR] [DOMAIN] [PROJECT_NAME]
```

Example:

```
./loop_variables.sh data-request.csv 2020 d02 EUROCORDERX
```

- The script reads variable names and frequencies from a CSV file
- One job per variable is submitted automatically
- Default settings assume SLURM (srun), but this can be modified to PBS or local execution in the shell script

Daily data aggregation:

Since the Fortran code does not generate daily output directly, daily mean, max, min, or sum fields can be created using `pCMORizer_aggregate_per_day.sh` code:

```
./pCMORizer_aggregate_per_day.sh [CSV_FILE] [PATH_to_1hr_FILES]  
[YEAR] [VARIABLE]
```

Notes

- CMORizer is still under development.
- If you encounter bugs or unexpected behavior, please report the issue in the GitLab repository ([link](#)).
- Make sure your software environment matches the requirements to avoid compiler or NetCDF conflicts.

Contact

For repository access:

Contact: Klaus G3rgen

Email: k.goergen@fz-juelich.de

For technical questions:

Contact: Klaus G3rgen and Josipa Milovac

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